

Voltage limit violation risk evaluation method considering communication failures in distributed voltage regulation system

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ABSTRACT

With the complexity and scale expansion of distribution network, the active operational management based on cyber physical system has been widely used in distribution network. Distributed voltage regulation is an advanced operational management approach, which requires reliable communication system. In order to evaluate the influence of communication failures on distributed voltage regulation, this paper proposes a voltage limit violation risk evaluation method considering communication failures. First, the model of distributed voltage regulation system is established and the consensus algorithm in distributed voltage regulation is detailed. On this basis, an improved consensus algorithm concerning the impact of communication failures is proposed. Second, a curve clustering algorithm is proposed to cope with the uncertainties of distributed generations. Then, a voltage limit violation risk evaluation method considering the uncertainties of communication failures and distributed generations is proposed. Finally, the proposed method is proved to simulate the distributed voltage regulation procedure considering communication failures. This study could provide a technical foundation for the planning and operation of active distribution network.

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1. Introduction

The rapid development of high-tech industries has put forward higher requirements for power quality. The uncertainty caused by the integration of a number of distributed generations (DGs) to the distribution network results in the bidirectional flow of the feeder, which increases the risk of voltage limit violations [1]. In order to suppress the negative impacts of gird-connected DGs, and to improve the power grid's capability to adapt to controllable resources, the distribution network is required to perform active operational management. Realizing the optimal control of DGs through collecting and processing in real-time power grid information can effectively enhance the economical and reliable operation of distribution network. While active control can effectively suppress the risk of power grid operation [2], it also brings new potential risk to the operation of the distribution network due to its deep dependence on the information system [3–6]. Therefore, it is profound to evaluate the operational risks considering communication failures, which can provide insights into control system design

and risk management. Table 1. Table 2. Table 3. Table 4..

One major challenge caused by gird-connected DGs is the voltage limit violation of DG connection node [7]. The voltage deviation of each node is usually maintained within the allowable range through voltage regulation in active distribution network (ADN). The state-of-art common voltage regulation strategies includes three main modes, decentralized control, centralized control and distributed control [8–11]. Due to the difference in control mechanism, the three modes have different requirements for information collection, information transmission and processing, which requires different degrees of dependence on communication system. The decentralized control methodology is the simplest approach, which only requires the local voltage as the input to realize voltage regulation. The voltage regulation device can absorb or inject power to decrease or increase the voltage of connection node. In this control methodology, voltage regulation devices don't need to communicate with each other [12]. Only the control law that computed by a centralized entity needs to be sent to the local controller. Therefore, the real-time response to the fluctuations can be realized whereas the

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Table 1

Communication device failure parameters.

Communication device	λ	Repair time (h)
Optical fibre	0.015f/km-a	5
Sensor	0.031f/a	2
Controller	0.031f/a	2
Agent node	0.005f/a	12

Table 2

Fibre optics length parameters.

Fibre Optics	Start-End nodes	Length (km)
LINE1/LINE2	22–25/25–22	18
LINE3/LINE4	22–33/33–22	18
LINE5/LINE6	18–33/33–18	14

Table 3

Voltage limit violation risk under various failure scenarios.

communication failure scenarios	Voltage limit violation risk
Controller	2.33 %
Sensor	0.52 %
Agent node	0.68 %
Fibre optics	0.58 %
Comprehensive risk	4.19 %

Table 4

Voltage limit violation risk under various failure scenarios.

communication failure scenarios	Voltage limit violation risk
Controller	6.05 %
Sensor	2.03 %
Agent node	0.61 %
Fibre optics	0.75 %
Comprehensive risk	9.44 %

global optimization control cannot be realized[13]. The centralized control methodology can achieve the best control performance. The control laws are computed by the common control center responsible for gathering system information according to some optimization objectives and constraints, and are sent to the remote devices via communication links[14]. The centralized control can achieve globalized optimization and is popular for small scale networks where power flow is unidirectional[15]. However, this control methodology requires an extended cyber infrastructure support to have an accurate knowledge about the whole network, which is merely available in distribution network[16].

Due to the increasing differences in the distribution network, the centralized control methodology is not suitable for all scenarios[17]. The distributed control methodology can reach the global optimization consensus in a distributed way with simple local communication, which overcomes the shortcomings of centralized and decentralized control strategy[18]. This methodology requires some agent nodes to receive and analyse information from one another and sensors surrounding them[19]. The agent nodes analyse and exchange status information with each other to achieve a global objective, which is usually realized by consensus algorithm[20–21]. Compared with centralized control, it reduces the dependence on the communication system and improves the economy and robustness of operation control. Thus, it has received more and more extensive attention in recent years. Reference [22] studied the distributed voltage control strategies of based on consensus algorithm. Different consistency variables were selected respectively according to control objects. The results showed that distributed control strategies can effectively reduce the risk of voltage limit violation and the configuration cost of communication system. Those studies mainly focused on the performance and algorithm of distributed control and

didn't involve the adversarial impact of communication failures. Although the distributed control methodology reduces the dependence on the communication system, necessary communication between the agent nodes and sensors is still needed to realize the optimal voltage control. The communication failures would make one or more DGs become unobservable and uncontrollable, which could cause a great increase in system operational risks, such as system control performance degradation[23], voltage deviation[24] and overcurrent[25]. The impact of cyber invalidity on centralized control performance was studied in [26]. However, the control principles of centralized control and distributed control methodology are different, which leads to the different impact of cyber invalidity.

In summary, with the distributed control methodology widely used in ADN, the adversarial impact of communication failures on system operational states should not be neglected. Thus, appropriately incorporating the influence of communication failures on distributed control performance is instrumental in devising operational strategies and guaranteeing the operational security of ADN. Focused on the ADN with distributed control strategies, this work proposes a risk evaluation approach for voltage limit violation in ADN with the consideration of communication failures. The major contributions of this paper are summarized as follows:

- 1) A curve clustering algorithm based on K-means method is proposed to cope with the uncertainties of DGs. The typical photovoltaic (PV) output curves are generated by the clustering algorithm to feature different operational scenarios with corresponding probabilities.
- 2) Analytical method based on accessibility matrix is adopted to cope with the uncertainties of communication failures. The accessibility coefficients that feature different communication failure scenarios are obtained from accessibility matrixes based on the status of communication devices to perform the voltage limit violation risk evaluation.
- 3) Based on the accessibility coefficient, the distributed consensus algorithm is improved so that the influence of communication failures can be coupled into the voltage regulation results. In this way, the node voltage profile under different failure scenarios can be obtained directly by power flow calculation.

The remainder of this paper is organized as follows. In Section 2, the model of distributed voltage regulation system and the consensus algorithm are analysed. The improved consensus algorithm considering communication failures is also detailed in this section. The modeling of uncertainties of DG outputs and the risk evaluation approach for voltage limit violation is introduced in Section 3. The case studies and simulation results are presented in Section 4. Finally, conclusions are drawn in Section 5.

2. Model of distributed voltage regulation system

2.1. Configuration and operation principle of distributed voltage regulation system

The distributed voltage regulation system (DVRS) consists sensors, controllers, fibre optics, agent nodes(ANs), photovoltaic inverters and battery energy storage systems(BESS), as shown in Fig. 1. Agent nodes refer to the work stations that composed of equipment with information relay and control action calculation capabilities. The status of local nodes is gathered by sensors and issued to the local AN. The local agent node exchanges the status with other neighbor agent nodes to generate the control instructions based on received information from local and neighbor nodes. Then, the control instructions are delivered to the local controller and the controller executes the received instructions to regulate the PV inverters and energy storage devices, thus realize the regulation of physical subsystem. The distributed regulation system usually adopts consensus algorithm, which has the advantages of simple

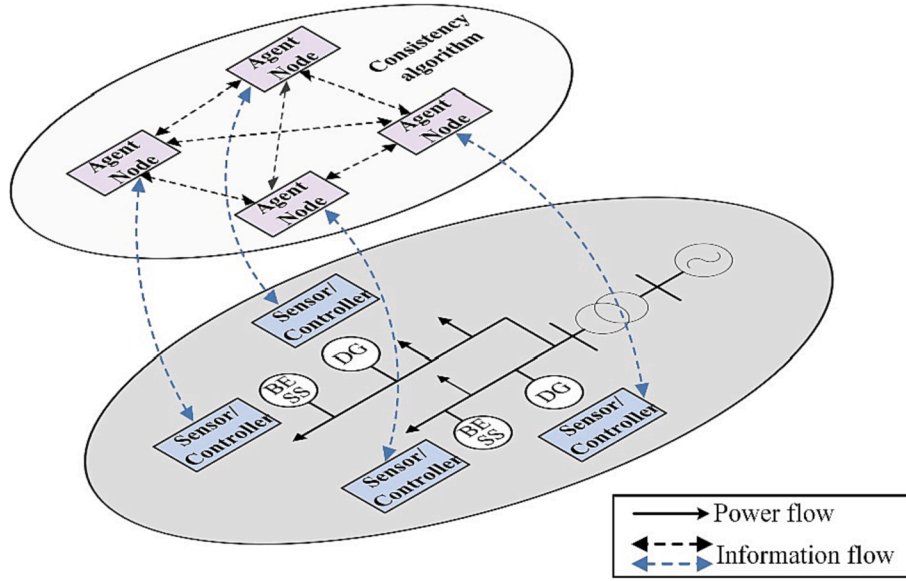


Fig. 1. distributed voltage regulation system.

modeling, fast convergence, and low configuration requirements for communication system.

2.2. Consensus algorithm in distributed voltage regulation

The basic principle of consensus algorithm is as follows: The consensus algorithm updates the control variables of the local nodes through the information interaction between the local node and the adjacent nodes. Through continuous iteration, the control variables of the nodes tend to be consistent. When the value of control variables of all nodes satisfy the iteration termination condition, the voltage regulation procedure is finished. Only the topology of communication system is a strongly connected graph, the consensus algorithm can converge, and the converged value is a constant value independent of the system topology.

The reactive power of the PV inverter is selected as the consistency variable, which can be described as $Q(k)$. The principle of consensus algorithm is expressed as follows:

$$Q_i^{(k+1)} = \sum_{j \in N_c} d_{ij} Q_j^{(k)} \quad (1)$$

$$|Q_i^{(k)} - Q_j^{(k)}| \leq 0.001$$

where k is the iteration times, d_{ij} is the transfer factor of the j -th agent node to the i -th agent node, which can be calculated by (2).

$$d_{ij} = \begin{cases} \frac{2}{n_i + n_j + 1} & j \in v_c(i) \\ 0 & \text{others} \end{cases} \quad (2)$$

where n_i and n_j represent the number of adjacent nodes of node i and node j , respectively.

The set of DG connection node is denoted as Θ_{PV} . When the voltage of these nodes exceeds the limit, the node whose voltage limit violation is recognized firstly is denoted as o . According to the consensus algorithm, the sensor of node o delivers the voltage value to the local agent node. If the nodal voltage exceeds the upper bound, the signal of absorbing reactive power will be issued to the controller from the agent node. If the nodal voltage exceeds the lower bound, then the signal of injecting reactive power will be issued to the controller from the agent node.

The control instruction is transmitted from node o to other nodes. The reactive power of PV inverter of node o can be calculated as (3).

$$Q_o^{(k)} = \begin{cases} -\frac{V_o^{(k)} - V^{\max}}{\sum_{j \in \Theta_{PV}} S_{o,j}^{V-Q}} & V_o^{(k)} > V^{\max} \\ 0 & V^{\min} < V_o^{(k)} < V^{\max} \\ -\frac{V_o^{(k)} - V^{\min}}{\sum_{j \in \Theta_{PV}} S_{o,j}^{V-Q}} & V_o^{(k)} < V^{\min} \end{cases} \quad (3)$$

where $S_{o,j}^{V-Q}$ is voltage-reactive power sensitivity coefficient, which means the voltage change of node o caused by per unit change of reactive power of node j . The calculation method is referred to Reference [20].

The distributed voltage regulation procedure is shown in Fig. 2, where VRN is the abbreviation of 'voltage regulation node'.

2.3. Distributed voltage regulation under communication failures

The distributed voltage regulation results will be different once the communication failures occur. The communication failures would affect the voltage control performance and cause increased operational risks. To appropriately incorporate the effects of communication failures, the consensus algorithm is modified to simulate the distributed control process in the event of communication failures. The regulation results could indicate the nodal voltage under communication failures.

(1) Uncertainties of communication failure scenarios

Communication failures of different device could result in different communication links losing availability as well as connection interruption. Thus, analysing the availability of communication links is the key to analysing the communication failure impact.

Currently, full duplex communication mode is mainly adopted in the communication system. The topology of communication system can be graphically represented as a directed graph, where the communication devices represented by the buses or nodes and the fibre optics are described as the edges. In this way, the validity of the communication link can be converted into the connectivity of the directed graph.

The directed graph is assumed as $D=(V,E)$, where the bus set is represented by $V=\{v_1, v_2, \dots, v_n\}$ while the edge set is represented by $E=\{e_1, e_2, \dots, e_m\}$. The directed edge, which starts from bus a and end with bus b , is defined as a_{ij} is used to represent the connectivity of bus v_i and bus v_j .

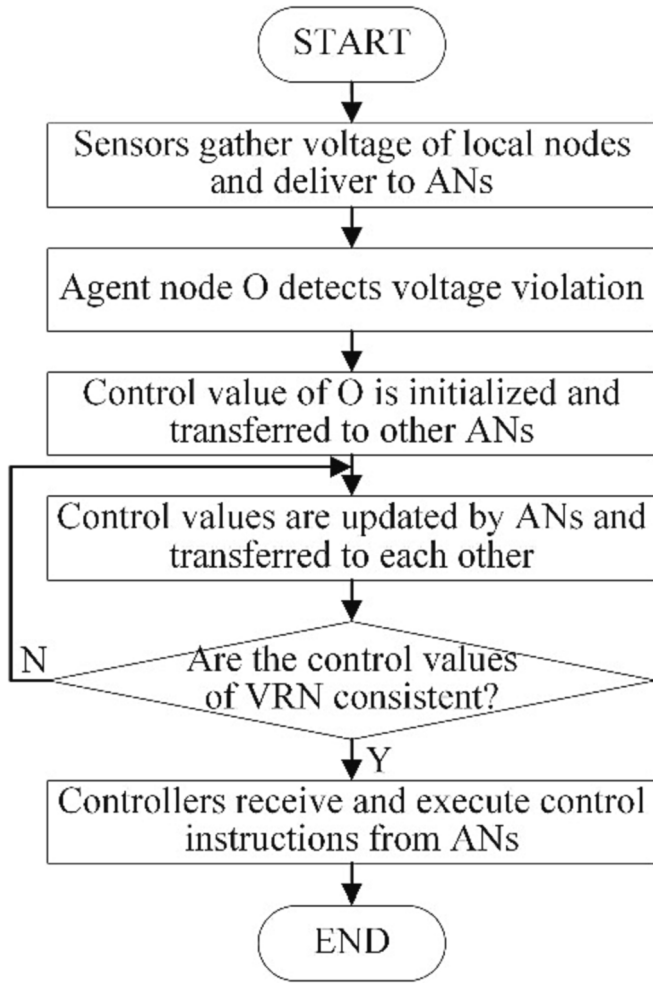


Fig. 2. Distributed voltage regulation procedure.

which can be calculated by (4). The total number of buses is n . The matrix $\mathbf{A} = \mathbf{A}(\mathbf{D})(n \times n)$ formed by a_{ij} is called adjacent matrix of directed graph $\mathbf{D}$.

$$a_{ij} = \begin{cases} 1 & \text{the directed edge starts from } v_i \text{ to } v_j \\ 0 & \text{others} \end{cases} \quad (4)$$

The matrix $\mathbf{T} = (t_{ij})_{n \times n}$ is defined as the accessibility matrix of directed graph $\mathbf{D}$. The element on the i -th row and j -th column can be expressed as (5).

$$t_{ij} = \begin{cases} 1 & \text{information can be transferred from } v_i \text{ to } v_j \\ 0 & \text{information cannot be transferred from } v_i \text{ to } v_j \end{cases} \quad (5)$$

The Accessibility Matrix $\mathbf{T}$ can be obtained from the adjacent matrix $\mathbf{A}$. The process of Accessibility Matrix calculation is illustrated as follows: 1) Calculate the matrix $\mathbf{B}_n$ by $\mathbf{B}_n = \mathbf{A} + \mathbf{A}^2 + \dots + \mathbf{A}^n$; 2) Modify the matrix $\mathbf{B}_n$: the non-0 elements are replaced by 1, and the elements equal to 0 remain the same. Then the accessibility matrix $\mathbf{T}$ can be obtained.

The failure scenarios are generated by the communication device failure enumeration. Enumerating the failed device out of N devices and obtaining the device statuses, the failure scenario and the corresponding scenario probability can be calculated by (6). It should be noted that only $N-1$ failures are considered in this paper.

$$P_{ff} = P_j \prod_{k=1, j \neq k}^N (1 - P_k) \quad (6)$$

where P_{ff} is the scenario probability of j -th failure scenario. N is the number of communication failure scenarios, which is equals to the

number of communication devices. P_j and P_k are the failure probabilities of the j -th and the k -th communication devices.

(2) Analysis of the influence of different communication device failures

Different communication devices play different roles in the voltage regulation. The information of physical subsystem is gathered and delivered to the agent nodes by sensors. The control instructions are generated and issued to the controllers by agent nodes. The controllers receive and execute the instructions to regulate the nodal voltage. And the upload and download of information rely on fibre optics.

The communication device with failure would be out of service during the repair time. This means that the information transmitted or processed by this device is invalid during the fault period. A sensor is an upstream device. If it fails, the local state information cannot be collected before the repair, and the node information cannot participate in the optimization calculation, which may lead to the inaccurate optimal control instructions. At the same time, it will also affect the control effect of other nodes. Instead, a controller is a downstream device, which only receives the information from agent nodes. When controller failures occur, the control instructions from the upper layer cannot be executed and the associated physical node remains in the same state during the fault period. Thus, only the control performance of local node is affected by the controller failure. For an agent node, which is considered as the brain of the communication system, the failure may cause that the control instruction for the associated controller cannot be generated and delivered. What's more, the information required by other agent nodes for optimization calculation is lost, which leads to the control instructions inaccurate. When the fibre optic failure occurs, the information transmitted via the fibre optic is invalid. The impact analysis is the same as above and no more detailed here.

(3) Consensus algorithm under communication failures

Most communication device failures in the communication system are repairable regardless of ageing failures. The element state can be simulated by the "run-stop-run" cycle. Thus, the state $S(k)$ of the communication device k can be expressed as (7).

$$S(k) = \begin{cases} 1 & \text{normal operation} \\ 0 & \text{contingency outage} \end{cases} \quad (7)$$

According to the method detailed in Section 2.3, the adjacent matrix of the communication system can be obtained. The information transmission interruption caused by communication link failures such as the ageing failure and mechanical broken of fibre optics can be considered as the unavailability of the directional edges, which is equivalent to setting the corresponding elements of the adjacent matrix as 0. The outages of sensors, controllers and agent nodes can be considered as the unavailability of the buses, which is equivalent to setting the corresponding elements of the adjacent matrix as 0. Then, the accessibility matrix could be obtained from the modified adjacent matrix. The sensor, agent node and controller of voltage regulation node (VRN) i are treated as a whole, which is called Intelligent Agent i . The evaluation procedure of nodal voltage of the system considering the communication failures is illustrated as follows.

1) Number the devices of Intelligent Agent i . The sensor, agent node and controller are numbered $3i-2$, $3i-1$ and $3i$, respectively. The topology of communication system is represented by adjacent matrix.

Taking the communication system shown in Fig. 3 as an example, the generation method of adjacent matrix and accessibility matrix is as follows.

The adjacent matrix can be expressed as Fig. 4.

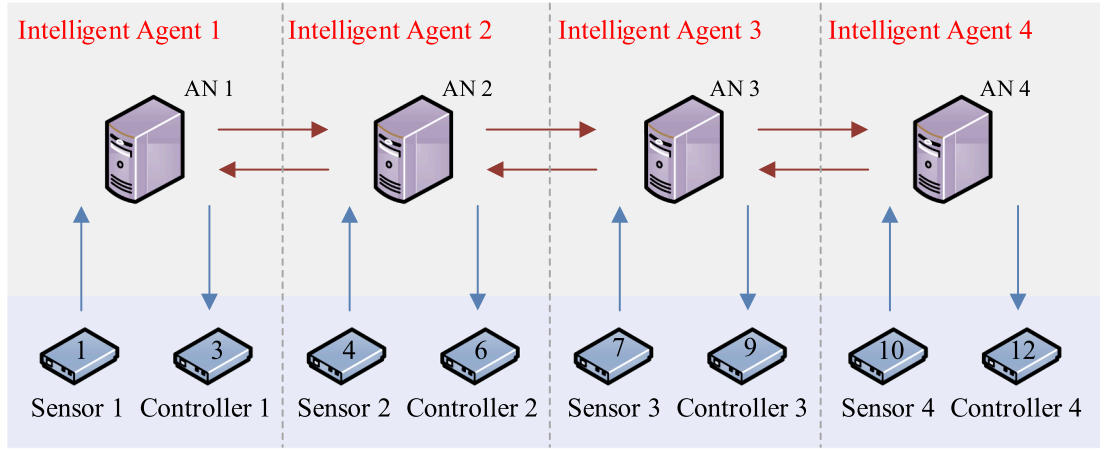


Fig. 3. 4-node communication system topology.

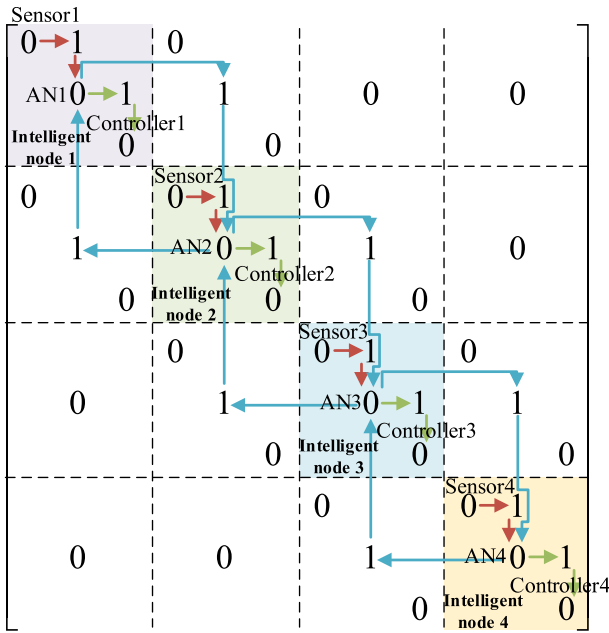


Fig. 4. Adjacent matrix of 4-node communication system.

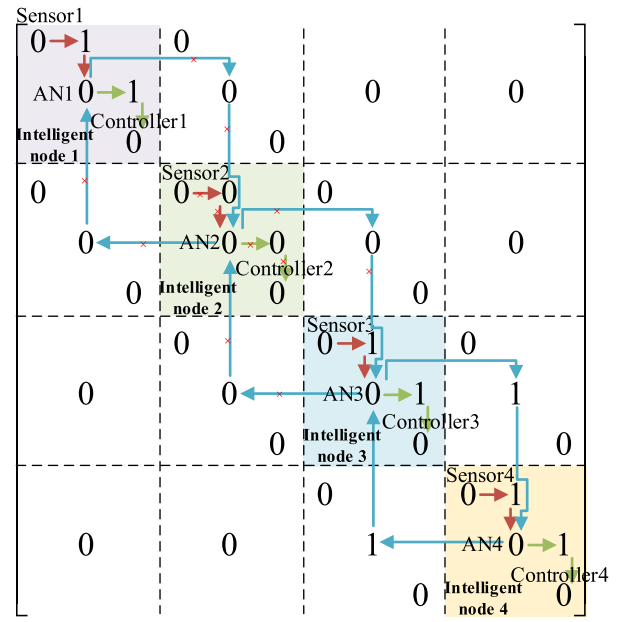


Fig. 5. Adjacent matrix with the agent node 2 being out of service.

a_{ij} is the element of the adjacent matrix. $a_{ij} = 1$ means that the information can be delivered from device i to device j directly. If the agent node 2 becomes out of service, the adjacent matrix should be modified as Fig. 5.

- 2) Based on the method detailed in Section 2.3, the accessibility matrix T can be obtained from the adjacent matrix A . According to the definition of adjacent matrix element a_{ij} and transfer factor d_{ij} , the equation as (8) can be obtained.

$$d_{ij} = \begin{cases} \frac{1}{\sum_{k=1}^n a_{(3k-1)(3i-1)} + 1} & i \in N_c(i) \\ 0 & \text{others} \end{cases} \quad (8)$$

where n is the number of nodes.

The accessibility matrix T with the agent node 2 being out of service can be expressed as Fig. 6.

t_{ij} is the element of the accessibility matrix T . $t_{ij} = 1$ ($i \neq j$) means that information can be delivered from device i to device j via communication devices. The dimension of transfer factor matrix is same with the

number of agent nodes. When the agent node 2 is out of service, transfer factor matrix D can be expressed as (9).

$$D = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix} \quad (9)$$

- 3) The consensus algorithm considering communication failures can be derived according to the transfer factor matrix D , accessibility matrix T and the control value $Q_0^{(1)}$ of the dominant node calculated by (3). The modified consensus algorithm considering communication failures can be expressed as (10).

$$Q_i^{(k+1)} = t_{(3i-1)(3i)} \sum_{j=1}^n t_{(3j-2)(3i-1)} d_{ij} Q_j^{(k)} \quad (10)$$

where t_{ij} and d_{ij} are the corresponding elements of the accessibility

Sensor1									
1	1	1							
1	1	1	0	0	0				
1	1	1							
Sensor2									
			0						
0			0	0	0				
			0						
Sensor3									
				1	1	1	1	1	1
0	0			1	1	1	1	1	1
				1	1	1	1	1	1
Sensor4									
				1	1	1	1	1	1
0	0			1	1	1	1	1	1
				1	1	1	1	1	1

Fig. 6. Accessibility Matrix T with the agent node 2 being out of service.

matrix and transfer factor matrix, respectively. $Q(k)j$ is the output value of the j -th controller. n is the number of voltage regulation nodes. The convergence condition is shown as (1) and the calculation method of control value of the initial node is shown as (3). The nodal voltage can be obtained by power flow calculation based on the reactive power control value.

3. Evaluation method of voltage limit violation risks

In order to observe the operational state of the system with communication failures under the distributed voltage regulation, a voltage limit violation risk evaluation method concerning the impact of communication failures is proposed in this section.

3.1. Uncertainties of PV output

According to the risk theory, the operational risk is caused by uncertainties. The uncertainties of PV output increase the risk of voltage limit violation. In order to appropriately evaluate the voltage limit violation risks of distributed control system, it is necessary to analyse and model the uncertainties of PV output to generate the operational scenarios.

The uncertainty of PV output could cause the voltage fluctuation. The greater the voltage fluctuation deviates from the rated value, the greater the necessity of voltage regulation, the heavier the dependence of physical system on communication system. What's more, the temporal correlation of PV output should be considered in the evaluation process due to the outage duration induced by communication device failures. Therefore, this paper proposes an operational scenario generation method based on curve clustering algorithm. The clustering is performed based on the historical data of annual PV output. In this way, the typical operational scenarios are generated, which are represented by some PV output curves and the corresponding probabilities.

(1) Operational scenario generation

The matrix which represent the set of PV output samples can be expressed as (11).

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1u} \\ x_{21} & x_{22} & \dots & x_{2u} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nu} \end{bmatrix} \quad (11)$$

where the i -th row of the matrix refers to the PV output samples of i -th day, which can be represent as follows:

$$X_i = [x_{i1}, x_{i2}, \dots, x_{iu}] \quad (12)$$

where x_{it} is the PV output value at time t .

The Euclidean distance is adopted to describe the difference between samples. The Euclidean distance of Sample X_i to X_j can be calculated by (13).

$$d_{ij} = \sqrt{\sum_{t=1}^u (x_{it} - x_{jt})^2} \quad (13)$$

where x_{it} and x_{jt} are the t -th data in the Sample X_i and X_j , respectively. u is the dimension of the sample. The smaller d_{ij} is, the more similar X_i and X_j are. All samples are divided into the clusters to which the corresponding samples have the smallest Euclidean distance.

The arithmetic mean of all samples in each cluster is calculated as the new cluster centre. The Euclidean distance from new cluster centres to all samples is calculated. Then repeat the above steps until the cluster centres no longer change. That is to say, Minimum Squared Error(MSE) reaches its minimum and no longer changes, which can be calculated by (14).

$$E = \sum_{r=1}^K \sum_{x_{(r)} \in C_r} (X_{(r)}^{(i)} - \bar{X}_{(r)}) (X_{(r)}^{(i)} - \bar{X}_{(r)})^T \quad (14)$$

where C_r is the r -th cluster. $X_{(r)}^{(i)}$ is the i -th PV output sample of C_r . $\bar{X}_{(r)}$ refers to the cluster centre of C_r . K is the number of clusters.

K of PV output samples are selected as the cluster centres, which can be denoted as $\bar{X}_{(1)}, \bar{X}_{(2)}, \dots, \bar{X}_{(K)}$. All of the samples with the number of n are divided into k clusters. The K clusters can be expressed as $C_1, C_2, \dots, C_K$. the number of samples of C_r is denoted as a_r , we have: $a_1 + a_2 + \dots + a_K = n$. $X_{(r)}^{(i)}$ and $\bar{X}_{(r)}$ are u -dimensional vector. The sums of squared deviations of C_r can be expressed as (15).

$$W_r = \sum_{i=1}^{a_r} (X_{(r)}^{(i)} - \bar{X}_{(r)}) (X_{(r)}^{(i)} - \bar{X}_{(r)})^T \quad (15)$$

ξ^2 is adopted in this paper to describe the increment in the sums of squared deviations after the two clusters are merged into a new cluster, so as to determine the difference of the two clusters and whether the two clusters should be merged. ξ^2 is expressed as (16).

$$\xi^2 = \frac{W_s - (W_z + W_r)}{(W_z + W_r)/(a_z + a_r - 2)} \quad (16)$$

where W_s , W_z and W_r refer to the sums of squared deviations of cluster C_s , C_z and C_r . C_s is the new cluster merged by C_z and C_r . a_z and a_r are the number of samples of C_z and C_r , respectively.

The number of initial cluster centres is K . The K clusters are merged until K equals to 1. The relationship between K and ξ^2 is recorded and the increment of ξ^2 is calculated by (17).

$$\gamma_{(K,K-1)} = \frac{\xi_{(K-1)}^2}{\xi_{(K)}^2} \quad (17)$$

Different clustering accuracy requires different γ value which is set as the merging criteria. In this paper, it is considered that when γ is greater than 10, the difference between the two clusters is significant. That is to say, the optimal number of clusters is the number of clusters of the last time merging with $\gamma < 10$.

(2) Operational scenario probability

The cluster centre of each cluster is considered as a typical operational scenario. the probability of the operational scenario can be represented by (18).

$$P_i = \frac{n_i}{365} \quad (18)$$

where n_i is the number of samples of the i -th operational scenario.

3.2. Voltage limit violation risk evaluation

The voltage limit violation risk is defined as the product of the probability of voltage exceeding the allowed ranges and its consequences. Analytical method is adopted in this paper to estimate the voltage limit violation risk. The risk value can be calculated by (19).

$$Risk = \sum_{t=1}^{\mu} \sum_{i=1}^K P_i \sum_{j=1}^N P_{fj} \sum_{\tau=1}^{96} S_{(t,\tau)}^{(ij)} \quad (19)$$

where Risk is the comprehensive risk value of voltage limit violation. P_i is the probability of the i -th PV operation scenario calculated by (18). P_{fj} is the probability of the j -th communication failure scenario calculated by (6). $S_{(i,j)}(t,\tau)$ is the value of the voltage out of limit of t -th node at time τ under the j -th communication failure scenario as well as the i -th operation scenario. μ is the number of nodes of the distribution network. K is the number of operation scenarios. N is the number of the communication failure scenarios.

$S_{(i,j)}(t,\tau)$ can be expressed as (20).

$$S_{(i,j)}^{(ij)} = \begin{cases} \frac{V_{(t,\tau)}^{(ij)} - 1.05}{1.05} & V_{(t,\tau)}^{(ij)} > 1.05 \\ 0 & 0.95 < V_{(t,\tau)}^{(ij)} < 1.05 \\ \frac{0.95 - V_{(t,\tau)}^{(ij)}}{0.95} & V_{(t,\tau)}^{(ij)} < 0.95 \end{cases} \quad (20)$$

where is the voltage value of t -th node at time τ under the j -th communication failure scenario as well as the i -th operation scenario.

The procedure of voltage limit violation risk evaluation is detailed in Fig. 7.

4. Case studies

4.1. Test system structure and parameters

A modified IEEE 33-node distribution system is used to perform case studies in this paper. There are 8 PV units connecting to the distribution network. The node 18, 22, 25 and 33 with PV connected are set as the voltage regulation nodes. The test system configuration is shown in Fig. 8. The detailed communication device failure parameters are listed in Tab.1. The fibre optics length parameters are listed in Tab.2. The outputs of the 8 PV units can be considered as equal since the DG connection nodes are not far away from each other. The historical DG output data of 365 days with the sampling rate of 15 min is used in this paper. In addition, only the single failure of communication devices is

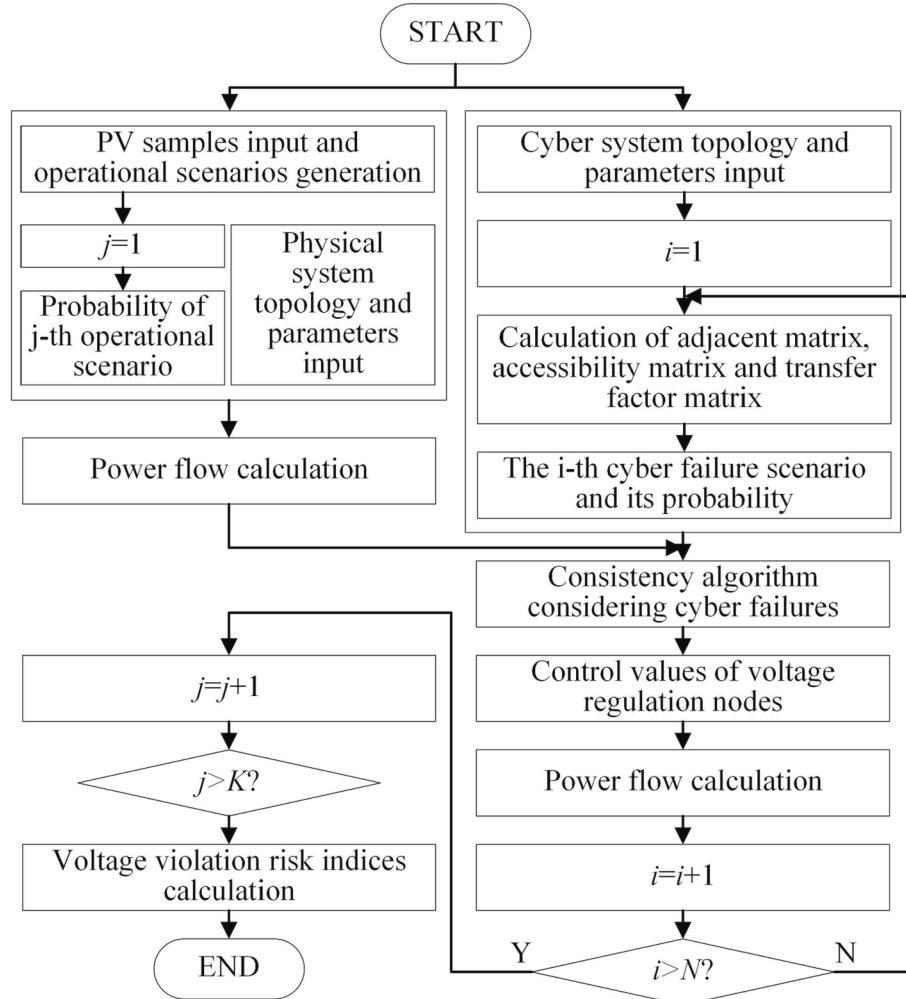


Fig. 7. Procedure of voltage limit violation risk evaluation.

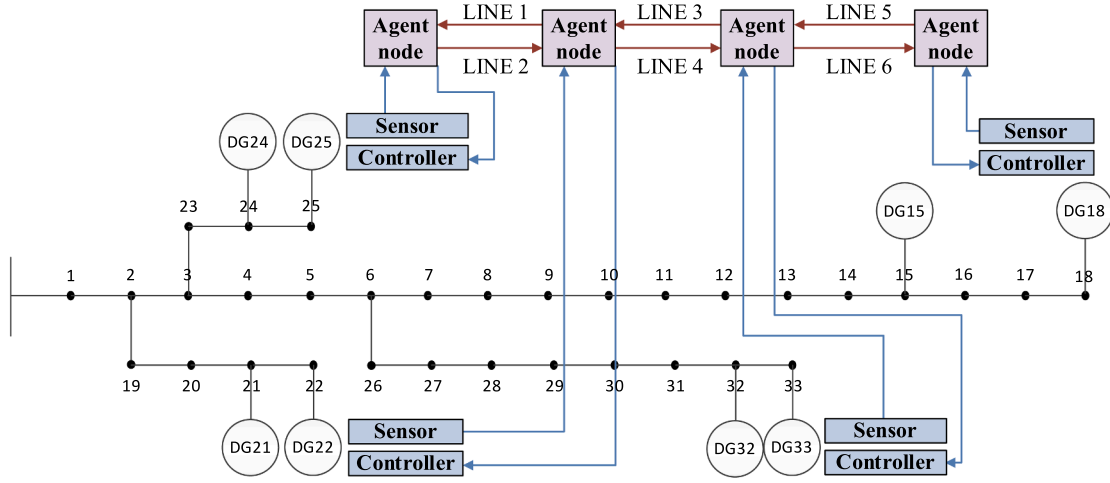


Fig. 8. Test system topology.

considered in this paper since the probability of multiple failures in the communication system is quite low.

4.2. PV output clustering results

There are 6 operational scenarios obtained by curve clustering. The cluster centres are shown in Fig. 9.

4.3. Voltage limit violation risk evaluation

Based on the proposed evaluation method of voltage limit violation risk considering communication failures, the voltage limit violation risks in case of various communication device failures are simulated and listed in Tab.3.

The node voltage profile under different failure scenarios is shown in Fig. 10.

It can be seen from Tab.3 and Fig. 10 the voltage limit violation caused by the failure of the agent node is the most serious, and the voltage limit violation risk caused by the failure of the controller is the largest. This is because that the agent node is the brain of data-

transmission, computation and decision-making, whose fault would lead to the invalidity of local sensor, controller, as well as the connected fibre optics. Thus, the resulting control commands of local and neighbour nodes will be inaccurate. The controller failure only affects the control performance of local voltage regulation node. The voltage limit violation of local node can be mitigated by the reactive power injection of the neighbouring nodes, which results in lower voltage deviation. However, the voltage limit violation risk is higher due to the high failure probability of controller.

4.4. Comparison of failures of agent nodes in various locations

In order to analyse the difference in the impact of the same type of communication device failures in different locations, the modified distributed voltage regulation procedure is simulated and the results are shown in Fig. 11.

It can be seen from Fig. 11 that the failure of agent node 22 has the greatest impact on the control performance. This is because that agent node 22 serves as the pivot node among all agent nodes. The neighbouring agent nodes will lose contact once the failure of agent node 22

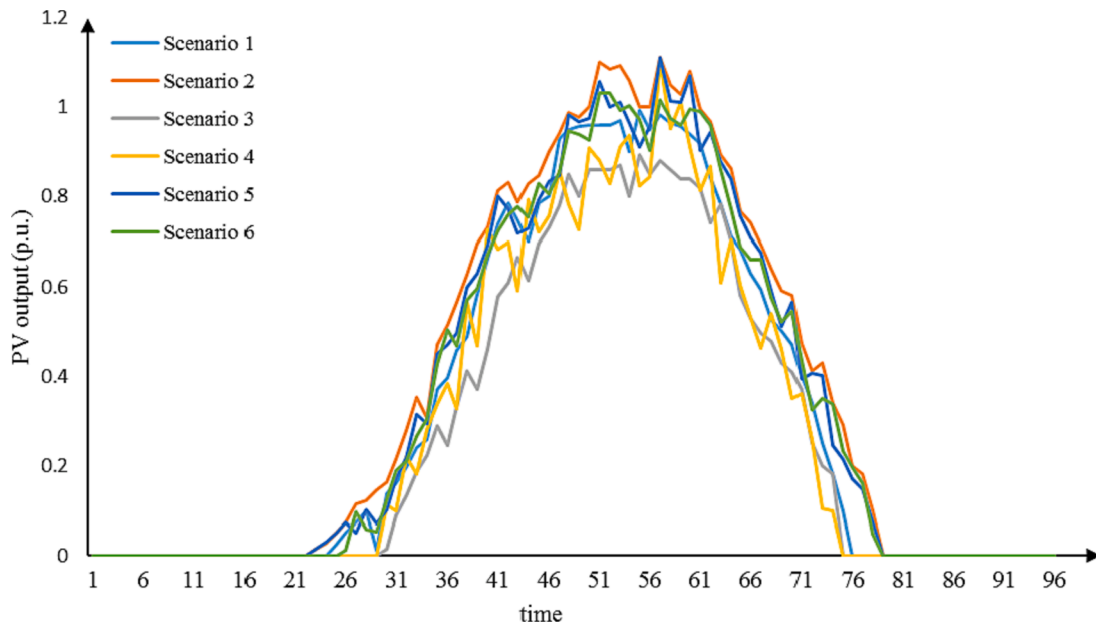


Fig. 9. Operational scenarios after clustering (cluster centers).

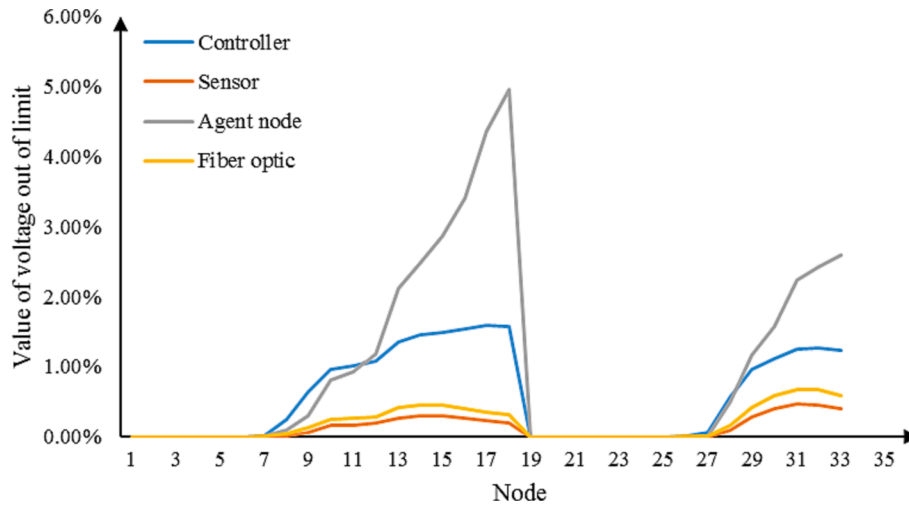


Fig. 10. Node voltage profile under various failure scenarios.

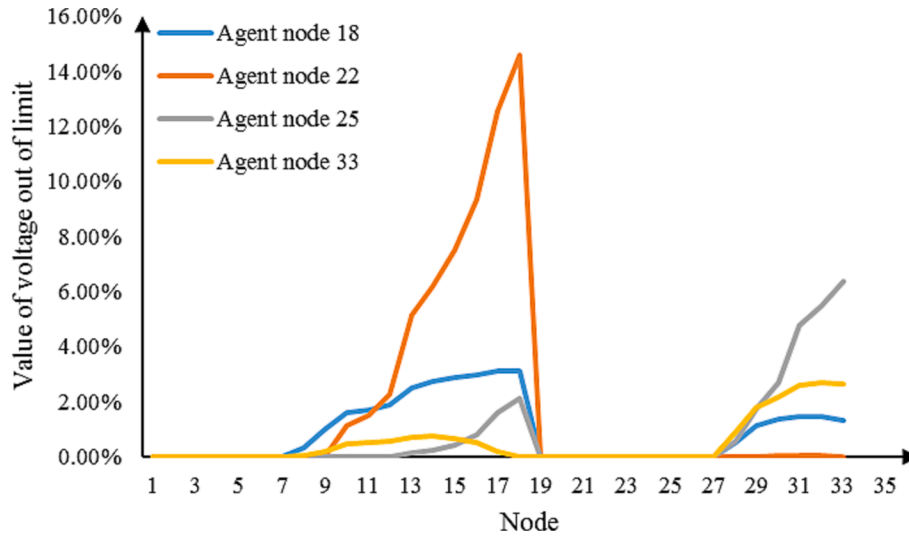


Fig. 11. Node voltage profile under various agent node failures.

occurs, which results in islanding operation of node 18. Thus, the initial value of the distributed control without iterations is used as the reactive power control amount, which lead to severe voltage limit violation.

4.5. Comparison of different configuration of voltage regulation nodes

In order to study the impact of voltage regulation node locations on the control performance, the modified distributed voltage regulation procedure is simulated for different voltage regulation nodes. Based on the test system, the nodes of 15, 21, 24 and 32 are chosen as voltage regulation nodes in this section. The voltage limit violation risks caused by different communication devices are shown in Tab.4.

Compared to the communication system configuration scheme in Section 4.1, the voltage limit violation risks are greatly increased in this section. This is because the source-load unbalanced nodes are chosen as voltage regulation nodes. Although this has a good voltage regulation effect, it requires the information system to be stronger and more reliable. In the event of a communication failures, the DGs' random power output could cause severe voltage fluctuations. Therefore, the choice of voltage regulation nodes should vary according to different applications. For better voltage regulation performance, the source-load unbalanced nodes are good choices as the voltage regulation nodes, which would be

with higher voltage risk. For the scenario with strict risk management and control, the source-load balanced nodes should be selected as the voltage regulation nodes, which ensures that the voltage risk is at a low level by sacrificing the performance of voltage regulation.

5. Conclusions

Considering the impact of communication failures, this paper proposed a new voltage limit violation risk assessment method, which simulates the distributed voltage regulation procedure under different communication failures. According to the simulation results, it can be concluded that:

- 1) The failures of agent nodes have the greatest impact on the control performance, especially the pivot node failure, which results in the most severe voltage limit violation.
- 2) The failures of controllers only affect their local or a few nearby nodes. For the nodes with large source-load imbalance, the voltage limit violation will exceed the limit to a much greater extent.
- 3) Because of the different degrees of source-load balance, the voltage limit violation risks brought by communication failures are different when selecting different nodes as voltage regulation points.

However, the overall risk of distributed control is less than the centralized control. The communication network can be configured according to the distribution of the source and load of the ADN, so as to ensure that the voltage limit violation risk is within an acceptable range on the premise of ensuring a good voltage regulation effect.

With the extensive research and further application of distributed control, it will help to improve the utilization efficiency of distribution network assets.

CRedit authorship contribution statement

Yu Lu: Methodology, Formal analysis, Software, Writing – original draft. **Mengdi Fu:** Data curation, Writing – original draft. **Jin Gao:** Investigation, Visualization. **Qingxin Shi:** Supervision. **Wenxia Liu:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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